

TAUPO, New Zealand

WMO: 932450

Lat: 38.742S

Lon: 176.081E

Elev: 402

StdP: 96.59

Time Zone: 12.00 (NZL)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-2.3	-1.2	-3.7	2.9	-1.3	-2.7	3.2	-0.2	12.3	9.0	11.2	8.7	1.6	80	0.536

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	9.4	24.9	17.6	23.6	17.1	22.4	16.7	19.4	22.6	18.6	21.7	17.8	20.8	3.5	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.3	13.9	20.9	17.4	13.1	20.2	16.5	12.4	19.5	57.1	22.7	54.3	21.7	51.7	20.8	22.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.0	9.7	8.6	DB	-4.2	27.5	0.5	1.3	-4.5	28.5	-4.8	29.2	-5.1	30.0	-5.5	31.0	(4)
(5)			WB	-4.5	20.6	0.6	0.8	-4.9	21.2	-5.3	21.7	-5.6	22.1	-6.0	22.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.8	17.2	17.3	15.4	12.2	9.9	7.6	6.8	7.4	9.1	11.0	12.9	15.5	(6)	
	DBStd	4.47	2.51	2.48	2.47	2.67	2.95	3.01	2.68	2.38	2.33	2.41	2.60	2.44	(7)	
	HDD10.0	389	0	0	1	9	39	83	104	84	44	19	6	1	(8)	
	HDD18.3	2413	51	44	94	184	261	322	358	338	277	227	163	91	(9)	
	CDD10.0	1054	223	204	167	75	37	11	4	4	17	49	93	172	(10)	
	CDD18.3	37	16	15	2	0	0	0	0	0	0	0	1	3	(11)	
	CDH23.3	137	73	45	5	0	0	0	0	0	0	0	4	10	(12)	
	CDH26.7	8	5	2	0	0	0	0	0	0	0	0	1	0	(13)	
(14) Wind	WSAvg	3.6	3.8	3.4	3.4	3.2	3.3	3.5	3.4	3.3	3.7	4.2	4.2	3.8	(14)	
(15) Precipitation	PrecAvg	1268	91	87	89	98	106	119	128	125	109	104	94	117	(15)	
	PrecMax	1717	290	235	190	197	178	216	294	233	192	206	179	205	(16)	
	PrecMin	986	27	17	36	31	33	50	37	55	41	33	44	16	(17)	
	PrecStd	179	55	50	40	46	38	37	58	46	41	39	37	48	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.1	26.5	24.0	20.6	17.9	15.3	13.9	14.1	16.7	19.2	23.7	24.8	(19)	
		MCWB	18.6	18.6	16.8	16.3	15.2	13.7	12.1	11.4	12.6	13.1	15.7	17.2	(20)	
	2%	DB	25.2	24.7	22.4	18.9	16.6	13.9	12.6	13.0	15.2	17.5	21.2	22.8	(21)	
		MCWB	17.6	17.9	16.4	15.2	14.5	12.4	10.8	10.5	11.8	12.9	15.0	16.4	(22)	
	5%	DB	23.8	23.4	21.1	17.8	15.5	13.0	11.7	12.1	14.2	16.4	19.4	21.5	(23)	
		MCWB	17.3	17.4	15.9	14.7	13.6	11.4	9.9	9.8	11.4	12.4	14.3	15.9	(24)	
	10%	DB	22.4	22.2	20.0	16.8	14.6	12.1	10.9	11.3	13.3	15.3	17.8	20.2	(25)	
		MCWB	16.8	17.0	15.7	14.0	12.7	10.5	9.1	9.2	10.8	12.1	13.5	15.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.6	20.7	19.0	17.7	16.2	14.1	12.8	12.5	13.7	14.7	17.5	18.9	(27)	
		MCDB	24.5	23.6	21.3	19.2	17.1	14.9	13.7	13.4	15.2	17.4	21.0	22.1	(28)	
	2%	WB	19.5	19.5	17.9	16.3	15.0	12.7	11.2	11.1	12.7	13.8	16.1	17.8	(29)	
		MCDB	22.9	22.6	20.3	17.9	16.2	13.6	12.2	12.3	14.3	16.0	19.4	20.7	(30)	
	5%	WB	18.6	18.6	17.0	15.3	14.0	11.7	10.2	10.2	11.9	13.1	15.0	17.0	(31)	
		MCDB	22.0	21.7	19.7	17.2	15.3	12.7	11.4	11.6	13.8	15.5	18.3	19.9	(32)	
	10%	WB	17.7	17.8	16.3	14.4	12.8	10.7	9.4	9.5	11.1	12.4	14.1	16.2	(33)	
		MCDB	21.0	21.0	19.1	16.4	14.3	11.9	10.7	10.9	12.9	14.9	17.2	19.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.7	9.4	9.2	8.9	7.8	7.3	7.4	7.6	8.1	8.2	8.9	8.5	(35)	
		MCDBR	11.9	11.0	10.4	9.4	7.3	5.8	6.3	7.8	8.9	9.8	11.5	10.6	(36)	
		MCWBR	5.7	5.3	5.5	6.1	5.3	4.8	4.7	5.6	6.0	6.2	6.4	5.6	(37)	
	5% WB	MCDBR	9.0	8.2	7.8	6.9	6.0	5.0	5.2	6.0	7.1	7.8	9.8	8.1	(38)	
		MCWBR	5.1	4.6	4.8	5.0	4.7	4.5	4.3	4.7	5.3	5.5	6.1	4.9	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.326	0.327	0.306	0.294	0.281	0.276	0.274	0.283	0.307	0.317	0.316	0.329	(40)	
	taud		2.571	2.575	2.640	2.664	2.677	2.663	2.671	2.645	2.547	2.531	2.556	2.558	(41)	
	Ebn at Noon		1006	981	964	916	870	843	869	914	945	980	1010	1007	(42)	
	Edh at Noon		105	100	86	73	62	58	61	73	92	103	106	107	(43)	
(44) All-Sky Solar Radiation	RadAvg		6.44	5.56	4.52	3.15	2.15	1.70	1.93	2.63	3.61	4.78	5.78	5.97	(44)	
	RadStd		0.47	0.56	0.28	0.23	0.14	0.15	0.17	0.19	0.31	0.34	0.46	0.53	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	+0.82	+1.00	N/A	N/A	N/A	-50	-117	N/A	N/A	
(47) Regional (6 neighbors)		+0.27	+0.36	+0.57	N/A	N/A	N/A	-24	-91	+83	N/A	

Nomenclature: See separate page